

Cryptocurrency Volatility Forecasting with Applications in Trading

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Abstract

We explore the benefits of using intraday realized volatility (RV) commonality, and propose a novel non-parametric framework for forecasting one-day ahead intraday RV. Specifically, a series of machine learning (ML) models are trained under various training settings (single-asset, cluster-driven, cross-asset, and pooled), where commonality progressively enters the model dynamics as training schemes become more complex. Our findings support the superiority of models that leverage the cryptocurrency commonality over those that do not explicitly account for it, regardless of market regimes. Furthermore, RV forecasts produced by our proposed models are employed in a spillover effect analysis and in trading strategies that trade cryptocurrency options in our universe, yielding higher Profit & Loss (P&L) in comparison with the benchmark model. The source code of this project is available at github.com/edjanga/crypto_volatility_commonality.

Keywords: Cryptocurrency markets, Volatility forecasting, Machine learning

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1 Introduction

The literature on volatility forecasting has gauged increasing interest since the pioneering work of Engle (1982) and Bollerslev (1986), where autoregressive conditional heteroskedasticity (ARCH) and generalized autoregressive conditional heteroskedasticity (GARCH) models were introduced, respectively. A large amount of studies have employed and expanded upon those baseline models, and aimed to improve their forecast abilities by considering different aspects of volatility processes. This has led to numerous extensions of the GARCH model that build on its shortfalls, *e.g.* nonlinear asymmetric GARCH (NGARCH) where the leverage effect¹ is part of the dynamics captured by the model of Engle and Ng (1993), exponential GARCH (EGARCH) where standard non-negativity constraints on the parameters are lifted by Nelson and Cao (1992), GARCH-in-mean (GARCH-M) by Hong (1991) where the mean equation contains a heteroskedasticity term, and Glosten-Jagannathan-Runkle GARCH (GJR-GARCH) from Glosten et al. (1993) that accounts for asymmetry of returns in the volatility process equation.

This abundance in models proposed to describe volatility dynamics demonstrates a strong interest of the research community in this area. However, this interest is not only strong amongst researchers. Indeed, Natenberg (1994) advised option traders to be aware of volatility models, such as ARCH and GARCH, as they capture both mean-reverting and serial correlation characteristics of volatility for the volatility forecasting task, and offer them additional tools to price options, as considered by Iqbal (2018). Option traders represent a subset of investment professionals that heavily rely on volatility for building trading strategies and hedging risk. The very concept of volatility is preponderant in a large amount of practitioners' decision process whether for portfolio and risk management (see Grinold and Kahn (2000)) or trading (see Kakushadze and Serur (2018)). This significant interest from both the academic and professional communities indicates how crucial volatility forecasts are for any asset class, including cryptocurrencies.

The popularity of cryptocurrencies has skyrocketed since Nakamoto's electronic peer-to-peer alternative to cash (Nakamoto (2008)). Many coins and tokens have been created after Bitcoin (BTC). Rapidly, exchanges where one can swap one token for another arose and researchers have investigated dynamics that govern this new asset class ever since. Similarly to other conventional asset classes, volatility has become a major concern in cryptocurrencies. Despite the highly volatile environment usually associated with cryptocurrencies, financial institutions have started to offer clients exposure to this new asset class. *e.g.* BlackRock launched an BTC ETF in January 2024. Consequently, the need to understand volatility dynamics in cryptocurrencies is of paramount importance as more participants are exposed to those markets.

Main contributions. In this study, we use intraday RV to forecast 1D-ahead intraday RV and the resultant forecasts are tested out-of-sample to assess the benefits of modeling cryptocurrency volatility with commonality. The contributions of this paper are three-fold: Firstly, we propose a novel non-parametric framework that exploits cryptocurrency intraday RV commonality using a basket of twenty cryptocurrencies. Secondly, we provide clear evidence of the benefits of leveraging commonality in intraday volatility to forecast 1D-ahead intraday RV in cryptocurrencies by assessing model performance in relative and absolute terms and by performing a spillover effect analysis to understand the dynamics governing our crypto RV network. Lastly, we explore the

¹Phenomenon describing the tendency of negative returns to increase future volatility by a larger amount than positive returns of the same magnitude.

economic benefits of RV forecasts in a trading strategy in the nascent option crypto currency markets. To the best of our knowledge, this is the first time that such a study is performed in the cryptocurrency market.

Outline. This paper is organized as follows: Section 2 describes the current state of research in volatility forecasting and shows how this work contributes to the literature. Section 3 presents the data from our numerical experiments, and the presence of commonality in cryptocurrencies, which also serves as a building block for the methodology developed in Section 4. We test and compare the performance model forecasts in Section 5. A study of the economic benefits of our models is presented in Section 6. We conclude our discussion and suggest potential research directions in Section 7.

2 Related Work

Our present study investigates the benefits of **commonality** in cryptocurrency **volatility forecasting** using **ML techniques**. An overview of the current state of related research is provided in this section.

The literature on daily volatility forecasting is extensive. However, research on utilizing intraday volatility in the context of cryptocurrency market remains relatively sparse. Over the past decade, the increasing availability of high-frequency data has prompted researchers to integrate intraday movement components into models for forecasting volatility. Corsi (2009) introduced an innovative approach to modeling daily realized volatility (RV) with his heterogeneous autoregressive (HAR) model, which captures the long-term memory of the volatility process through lags over various horizons. This model has inspired several studies that modify the HAR model by adapting it to intraday scenarios or by incorporating multiple features to leverage information from other variables, such as prices. These analyses have typically focused on standard asset classes, *e.g.* Zhang et al. (2024) incorporated intraday components to examine commonality in U.S. equities RV, Ye et al. (2023) concentrated on the last half-hour RV forecasts in the Chinese gold futures market, and Wang and Yang (2009) investigated asymmetric volatility in foreign exchange pairs by modifying the HAR model to include leverage and price jumps as features describing volatility dynamics.

Extending the HAR model is not the sole method for modeling daily volatility in the literature. Score-driven time series frameworks have also been employed for this purpose. These methods have demonstrated strong capabilities in capturing nonlinearity in time series. Following this line of thought, Catania and Grassi (2022) proposed a model that uses daily data to forecast cryptocurrency volatility. Their study, which involved a large number of cryptocurrencies (606) categorized into Large Cap, Medium Cap, and Small Cap, incorporated long memory, leverage effects, and time-varying higher-order moments in volatility processes. However, the authors did not include any cross-terms in the volatility dynamics. This paper introduces RV commonality gradually through different training schemes and leverages intraday data and machine learning techniques to forecast 1D-ahead intraday RV. The findings demonstrate that incorporating long memory, asymmetries, and higher-order moments significantly improves volatility forecasts.

Buccheri et al. (2021) proposed modeling intraday log-prices via a multivariate local-level model with score-driven covariance matrices using high-frequency data, such as 1-second NYSE stock data. The phenomena of commonality and asset co-movement are well-documented in

non-cryptocurrency asset classes. Prior studies have empirically demonstrated evidence of co-movement in volatility across different frequencies: [Herskovic et al. \(2016\)](#) in equities, [Bollerslev et al. \(2018\)](#) in commodities, FX, equities, and fixed-income on a daily basis, and [Zhang et al. \(2024\)](#) in equities on an intraday basis. [Engle and Sokalska \(2012\)](#) advocated using pooled data to enhance daily volatility forecast performance. However, little attention has been paid to commonality and co-movement in cryptocurrency volatility. [Mensi et al. \(2019\)](#) examined commonality between BTC and five major tokens—ETH, Dash, Monero, Ripple, and Litecoin using the cross-wavelet transform, as applied by [Hudgins et al. \(1993\)](#) to atmospheric turbulence. [Katsiampa \(2019\)](#) studied volatility co-movement in BTC and ETH, finding evidence of interdependencies between the two cryptocurrencies. These gaps in the literature regarding commonality and co-movement in cryptocurrency intraday volatility are addressed in our current work.

Machine learning (ML) techniques have significantly improved solutions to classical financial tasks over recent years, often outperforming econometric models, including in volatility forecasting (see [Carr et al. \(2019\)](#)). Several ML algorithms, such as Least Absolute Shrinkage and Selection Operator (LASSO), Principal Component Regression (PCR), Random Forest (RF), and other ensemble methods, produced superior results compared to traditional techniques (see [Li and Tang \(2022\)](#)). Other areas of ML have also been explored to enhance volatility forecasts. For instance, [Moreno-Pino and Zohren \(2022\)](#) introduced DeepVol, a model based on Dilated Causal Convolution, to predict one-day-ahead intraday RV on high-frequency US equities returns. ML applications in cryptocurrency RV forecasting have also emerged in recent years. [D’Amato et al. \(2022\)](#) utilized Recurrent Neural Networks (RNN) to forecast RV using both daily and intraday volatility. [Aaltio et al. \(2022\)](#) aggregated BTC RV values using minute-sampled BTC returns over three-hour intervals and predicted future values based on past samples using various ML methods, including Multi-Layer Perceptron (MLP), Gated Recurrent Unit (GRU), Long Short-Term Memory (LSTM), Support Vector Machine (SVM), and Ridge Regression. Their analysis revealed that Ridge Regression produced the best forecasts, closely followed by MLP, GRU, and LSTM-based methods. These intraday forecasts were compared to predictions made by the Heterogeneous Autoregressive (HAR) model. [Wang et al. \(2023\)](#) made use of both RF and LSTM to forecast intraday RV of BTC, ETH, Litecoin (LTC), and XRP, using internal determinants, such as RV lags and previous trading information, as well as external determinants like technological, financial, and policy uncertainty factors. These forecasts were compared to predictions made by the GARCH model.

Many studies have been conducted on spillover effect analysis. [Iyer and Popescu \(2023\)](#) investigated this phenomenon for both returns and volatility amongst cryptocurrencies and various asset classes, *i.e.* equities, FX, fixed income, and commodities. They used a data set composed of 23 crypto asset daily price data series and 15 financial variables spanning the period 14th October 2015 to 1st January 2023. [Iyer and Popescu \(2023\)](#) analyzed spillover effect by utilizing the vector autoregressive (VAR) framework proposed by [Diebold and Yilmaz \(2012\)](#). Their VAR-based connectedness approach relies on forecast error variance decompositions (FEVD) and is used to assess the degree of interconnections amongst assets. [Iyer and Popescu \(2023\)](#) were able to look at spillover effect from a bidirectional perspective, *i.e.* portion of variance of asset i attributable to asset j and vice versa. This modulus operandi is invariant to the ordering in the VAR model, hence it is more robust. [Iyer and Popescu \(2023\)](#) found that the interconnections between crypto assets and financial assets are lower compared to spillovers among crypto pairs in terms of both sending and receiving returns and volatility. They showed that crypto assets primarily transmit spillovers to financial markets, although during periods of financial sector stress, the reverse may also oc-

cur. When cross-asset spillover occurred, evidence provided by [Iyer and Popescu \(2023\)](#) showed stronger interconnectedness between crypto assets and global equities, VIX, and gold, whereas spillovers with fixed income, FX, and other commodities were relatively moderate.

[Patel et al. \(2024\)](#) demonstrated the interdependencies between healthcare cryptocurrencies and traditional asset classes utilizing quantile VAR and wavelet quantile correlation on interday data. Once this interrelation had been proved, the authors were able to construct a diversified portfolio that achieved high risk-adjusted returns. [Cornell et al. \(2023\)](#) examined spillover effects in crypto markets by modeling hourly RV of 261 crypto pairs leveraging vector autoregressive (VAR) abilities. Rather than using [Diebold and Yilmaz \(2012\)](#) approach, they looked at node degrees as well as their associated distributions. Their findings highlighted complex causal networks among their crypto universe and the existence of leaders that influence the rest of the network members in price movements. Structural properties of nodes play a central role in our spillover effect analysis and we shall combine it with [Diebold and Yilmaz \(2012\)](#) approach to describe our crypto RV network before proving the economic benefits of our framework.

Although this paper investigates the same phenomenon, namely cryptocurrency RV forecasting, our study is distinguished in several key aspects. We explore intraday RV commonality using four distinct training schemes across 20 cryptocurrency pairs, whereas most prior studies either restricted their scope to fewer than 10 pairs (as in [Aaltio et al. \(2022\)](#), [Wang et al. \(2023\)](#), [D’Amato et al. \(2022\)](#), and [Mensi et al. \(2019\)](#)) or focused solely on daily RV (as in [Catania and Grassi \(2022\)](#)). Additionally, we perform a spillover effect analysis where we utilized the framework introduced by [Diebold and Yilmaz \(2012\)](#) and elements of graph theory to describe crypto RV network. Finally, we propose a trading strategy using crypto option chain data to demonstrate the economic benefits of our cryptocurrency volatility forecasts.

3 Data and Commonality in Cryptocurrencies

3.1 Data Description

Let $\log P_i$ represent the price process of asset i and assume $d\log P_{i,t} = \mu_i dt + \sigma_{i,t} dW_t$, where μ_i and $\sigma_{i,t}$ are the instantaneous drift and volatility processes of asset i respectively, and W_t denote a standard Brownian motion. RV is defined as

$$RV_{i,t}^{(u)} = \sum_{s=t-u+1}^t r_{i,s}^2, \quad (1)$$

$$r_{i,s} = \log P_{i,s} - \log P_{i,s-u}, \quad (2)$$

where we set u to 30 minutes in our analysis, i.e. we focus on the modeling of 30-min intraday RV. Our data set was retrieved from tardis.dev² and spans a two-and-half year period, namely from January 2021 to June 2023. It consists of RV observations that were resampled to 5 minutes in line with the literature (see [Liu et al. \(2015\)](#)). The cryptocurrency universe is composed of the twenty most liquid USDT pairs on Binance: 1INCHUSDT, AAVEUSDT, ADAUSDT, AVAXUSDT, BCHUSDT, BNBUSDT, BTCUSDT, DOGEUSDT, DOTUSDT, EO-

²<https://tardis.dev/>

SUSDT, ETHUSDT, FILUSDT, HBARUSDT, LINKUSDT, LTCUSDT, MATICUSDT, SOLUSDT, UNIUSDT, XLMUSDT, XRPUSDT. We drop the USDT suffix for the sake of brevity hereafter. Note that for any pairs' liquidity i , we use market depth (MD) as a proxy, where $MD = \sum_{k=1} \sum_{s=1}^T P_s^k V_s^k$ with k corresponding to k -th orderbook level.³ All pairs are traded against USDT which allows one to report results in US dollar (USD) as USDT is pegged to USD.⁴ Summary statistics of intraday RV patterns are shown in Figure 1 and evidence of strong correlation in our cryptocurrency universe is supported by the heatmap depicted in Figure 2.

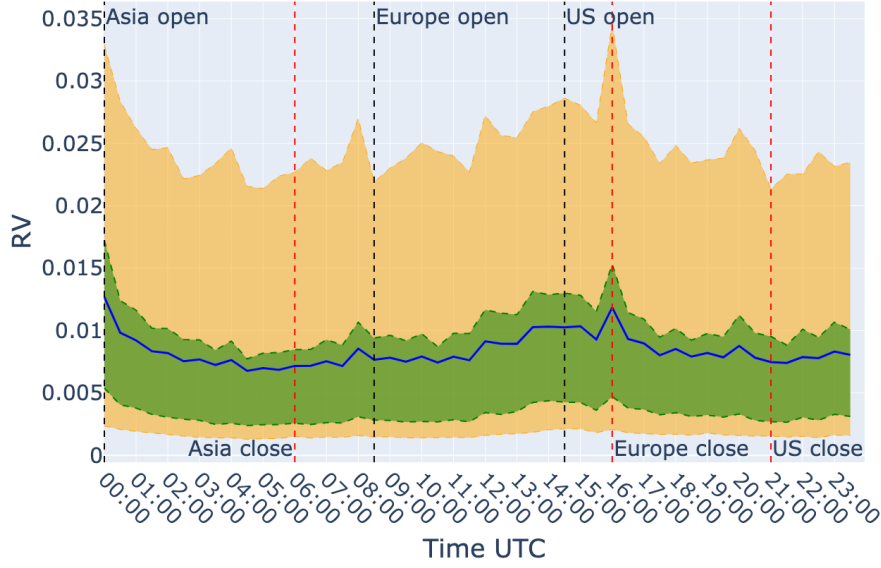


Figure 1: Annualized intraday RV statistics. Yellow and green areas delimit the (5^{th} - 95^{th}), and (25^{th} - 75^{th}) percentile bands, respectively. We overlay the opening (black) and closing (red) hours of three major equity markets: Tokyo Stock Exchange (Asia), London Stock Exchange (Europe), and New York Stock Exchange (US).

³Volume is defined as the sum of the volumes traded at the best bid and ask: $V = V^B + V^A$; here V^B (resp. V^A) stands for the volume at the best bid (resp. ask).

⁴We also winsorized the RV series. *Winsorisation* is a process that sets values higher (lower) than $(100 - \alpha)^{th}$ (α^{th}) percentile to that percentile. We opted for $\alpha = 1$ in this work.

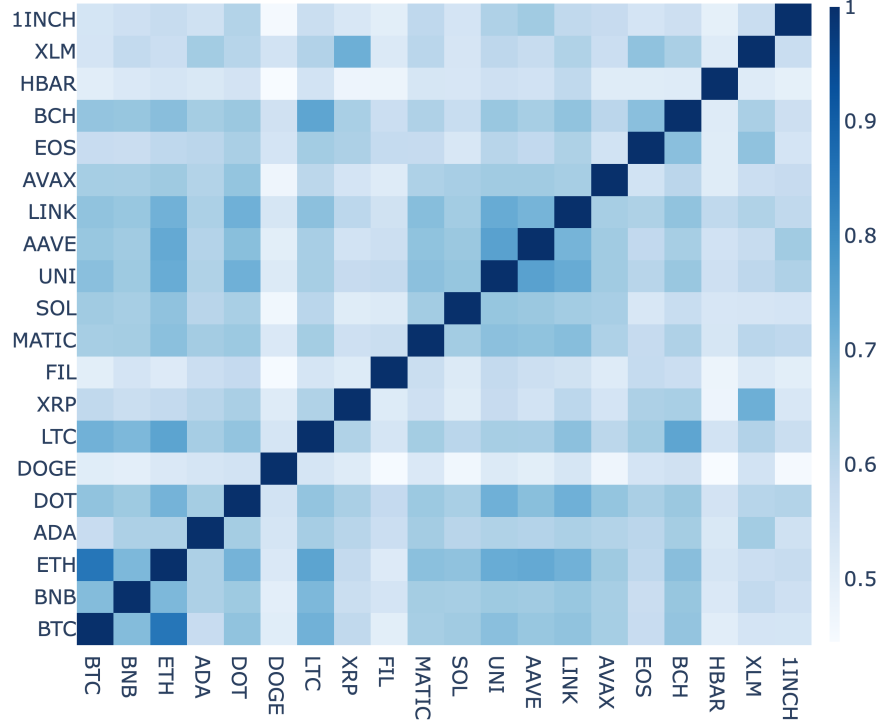


Figure 2: Daily pairwise RV correlation mean. Apart from DOGE which displays relatively low correlation with the rest of our basket, our cryptocurrency universe exhibits strong pairwise correlation.

Figure 2 hints the presence of commonality within our crypto pairs and we shall investigate it further in Section 3.2.

3.2 Commonality

In this section, we establish the presence of commonality in our crypto RV dataset via two different means. Similarly to Zhang et al. (2024), we showcase the evidence of commonality among cryptocurrency pairs thanks to a single-factor model for each pair, i.e.

$$RV_{i,t}^{(h)} = \alpha_i + \beta_i RV_{M,t}^{(h)} + \epsilon_{i,t}, \quad (3)$$

where $RV_{M,t}^{(h)}$ is the contemporaneous market volatility as defined by Bollerslev et al. (2018), i.e., equally-weighted mean of individual pair RVs. Commonality is defined as the cross-average of the adjusted coefficient of determination (R^2) of Eqn (3). Figure 3 shows commonality in our universe for three lookback windows, namely weekly, monthly, and biyearly henceforth referred to as $1w$, $1m$, and $6m$, respectively.

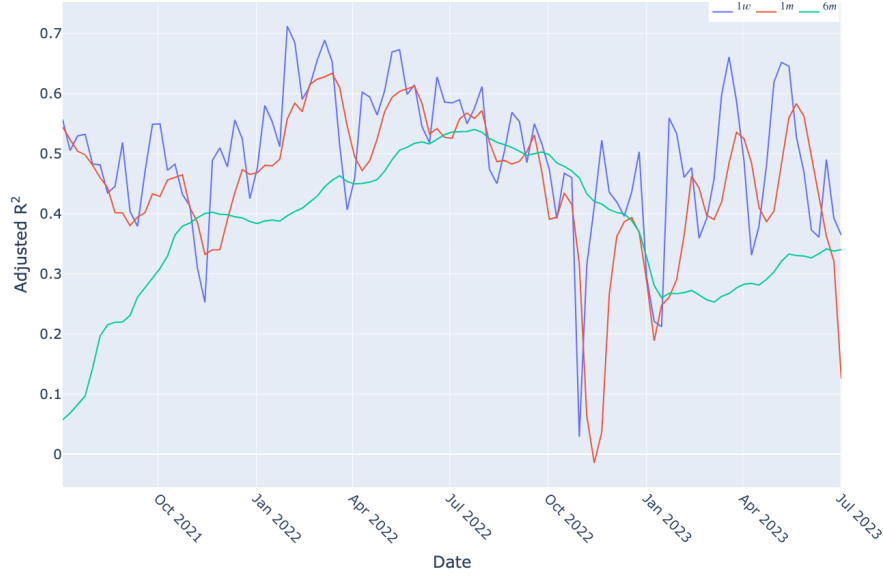


Figure 3: Commonality (type 1). The first type of commonality we look at is cross-pair commonality in realized volatility based on adjusted R^2 . We display commonality over the three aforementioned windows. L_{train} denotes the lookback window considered.

Furthermore, we make use of the Absorption ratio to describe intraday RV commonality in cryptocurrency markets. [Kritzman et al. \(2010\)](#) defined Absorption ratio as a measure of implied systematic risk describing to what extent markets are unified. This ratio is computed as the fraction of the total variance of a set of asset returns explained by a fixed amount of eigenvectors. We emulated the procedure proposed by [Kritzman et al. \(2010\)](#) for our crypto RV universe, we fixed the number of eigenvalues at $1/5^{th}$ of our universe, *i.e.* four eigenvectors. Figure 4 displays how the Absorption ratio evolves across time for all L_{train} values.

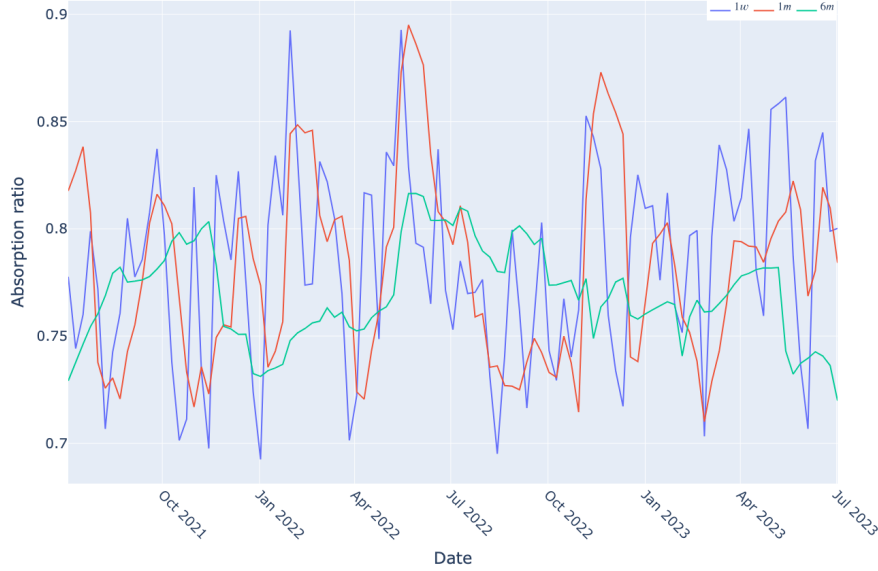


Figure 4: Commonality (type 2). We display cross-pair commonality in realized volatility based on absorption ratio. As hinted by this figure, the absorption ratio appears to decrease as L_{train} increases. Nevertheless, the score remains high which indicates that the market spanned by our crypto universe is intrinsically tied.

4 Methodology

4.1 Training procedure

In order to forecast 1D-ahead intraday RV, we train a model on a daily basis at midnight, keeping in mind that the cryptocurrency markets trade 24 hours a day. The size of the training sets depends on L_{train} . In other words, L_{train} controls the size of the training sets which can be 2016 ($=288 \times 7$) for $L_{train} = 1w$, 8640 ($=288 \times 30$) for $L_{train} = 1m$ or and 51840 ($=288 \times 180$) for $L_{train} = 6m$. While the training size is allowed to change, the size of the test set is fixed to 288 since one is interested in 1D-ahead intraday RV forecast. Figure 5 illustrates the training procedure.

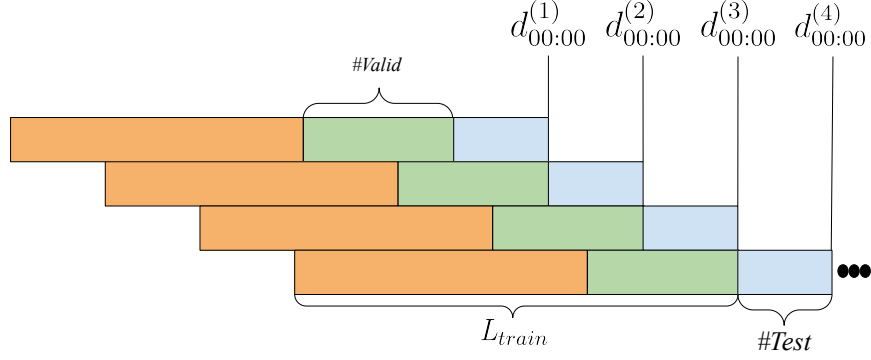


Figure 5: Training procedure. Amber and green boxes denote the training and the validation sets whose lengths depend on L_{train} , while blue boxes are the fixed test sets. Training sets represent 80% of L_{train} , and validation sets the remaining 20%. The use of validation sets allows for early stopping, which aims to speed up model training and hyperparameter tuning. For the remaining ML techniques, the training sets and L_{train} coincide. Each model is trained at midnight using the previous training set spanning a lookback period of length L_{train} and tested on the next day.

The parameter L_{train} defines the amount of RV lags used in each HAR model. Table 1 gives an overview of the RV lags in the models for distinct L_{train} values. Note that the lookback periods used to compute RV lags are capped by L_{train} , i.e. $L_f \leq L_{train}$, where L_f is the lookback period over which RV lag is computed as Table 1 suggests. RV lags have been grouped in two distinct categories as follows:

- *Dynamic lags (L^D)*: Each RV lag is built such that $RV_t^{(L_f)}$ is the RV mean of the L_f -rolling window, e.g. $RV_t^{(1H)}$ is the RV mean over the last one hour, or the 1-hour RV rolling mean.
- *Static lags (L^S)*: RV lags in this group are computed by taking the mean of RV over the previous fixed L_f -window, e.g. $RV_t^{(1w)}$ is the RV mean over the last seven days, and similarly $RV_t^{(1m)}$ and $RV_t^{(6m)}$ corresponds to the RV mean over the last 30 and 180 days respectively.

This training procedure is an instance of the walk-forward method (WF). De Prado (2018) stated that each decision, i.e. the coefficient estimates in our scenario, depends solely on the period predating that decision, i.e. L_{train} . He argued that WF benefits are two-fold: a clear historical interpretation, hence the performance of models trained with that method can be assessed over a historical time horizon, and the guarantee of performing testing out-of sample, thus avoiding data leakage. We elaborate more on the validation method employed in our experiments in Section 5.

Features								
	Dynamic lags					Static lags		
L_{train}	$RV_t^{(u)}$	$RV_t^{(1H)}$	$RV_t^{(6H)}$	$RV_t^{(12H)}$	$RV_t^{(1d)}$	$\overline{RV_t^{(1w)}}$	$\overline{RV_t^{(1m)}}$	$\overline{RV_t^{(6m)}}$
$1w$	•	•	•	•	•	•		
$1m$	•	•	•	•	•	•	•	
$6m$	•	•	•	•	•	•	•	•

Table 1: Summary of features for different lookback windows used to forecast $RV_{t+u}^{(u)}$ for HAR-based models.

4.2 Models

4.2.1 RiskMetrics

Since its introduction by [Longerstaey and More \(1995\)](#), RiskMetrics has been widely used for volatility forecasting purposes in industry. Evidence of GARCH in-sample superiority over RiskMetrics has been demonstrated, as mentioned by [McMillan and Kambouroudis \(2009\)](#). Nevertheless, this in-sample dominance does not necessarily translate to forecasting performance superiority ([McMillan and Kambouroudis \(2009\)](#)). This fact, coupled with the relative simplicity of RiskMetrics (see Eqn (4)), explain the model’s status in the investment community. Its popularity amongst practitioners has led multiple academic studies to include RiskMetrics for various goals and different settings, *e.g.* model assessment such as the out-of-sample evaluation conducted by [McMillan and Kambouroudis \(2009\)](#) or benchmarking with a group of other models, as carried out by [Symitsi et al. \(2018\)](#). In their initial work, [Longerstaey and More \(1995\)](#) formulated RiskMetrics in two ways: equally weighted (mean of demeaned squared returns series over a time interval T), and exponentially weighted (EW) forms. We fitted RiskMetrics using the EW form

$$RV_{i,t+u}^{(u)} = (1 - \lambda) + \sum_{s=1}^t \lambda^{s-1} (r_{i,s} - \bar{r}_i)^2, \quad (4)$$

where $r_{i,t}$ denotes the returns series of an asset i and $\bar{r}_i$ corresponds to its mean. The λ parameter corresponds to the rate of decay and [Symitsi et al. \(2018\)](#) recommend to set it to 0.94, as per consensus. [Symitsi et al. \(2018\)](#) list the lack of interpretability of λ as one of the shortfall of RiskMetrics, as no theoretical ground justifies the value λ .

4.2.2 Autoregressive

The dynamics of autoregressive (AR) are assumed by the following equation

$$RV_{i,t+u}^{(u)} = \alpha + \beta^{(u)} RV_{i,t}^{(u)} + \epsilon_{t+u}. \quad (5)$$

4.2.3 Heterogeneous autoregressive

As mentioned in Section 2, [Corsi \(2009\)](#) introduced a new class of models, dubbed as HAR, that aims to describe daily RV dynamics using RV lags over different time horizons. Despite its simplicity, multiple studies corroborated this statement by reporting good forecasting performance using daily data (see Section 1).

Similarly to [Zhang et al. \(2024\)](#), we modified the original HAR to allow our models to describe intraday movements. This was achieved by the addition of intraday RV lags, namely u , $1H$, $6H$, $12H$ lags where H stands for hours.

$$\begin{aligned} RV_{i,t+u}^{(u)} = & \alpha + \beta^{(u)} RV_{i,t}^{(u)} + \beta^{(1H)} RV_{i,t}^{(1H)} + \beta^{(6H)} RV_{i,t}^{(6H)} \\ & + \beta^{(12H)} RV_{i,t}^{(12H)} + \beta^{(1d)} RV_{i,t}^{(1d)} + \beta^{(1w)} \overline{RV_{i,t}^{(1w)}} \\ & + \beta^{(1m)} \overline{RV_{i,t}^{(1m)}} + \beta^{(6m)} \overline{RV_{i,t}^{(6m)}} + \epsilon_{t+u} \end{aligned} \quad (6)$$

This model will serve as the basis for the following models, and we shall refer to it as HAR throughout the rest of this paper.

4.2.4 Heterogeneous autoregressive with equity market session effects

Similarly to the study of [Liu and Serletis \(2019\)](#), we investigated the spillover effect of major equity markets on cryptocurrency RV via binary features. For simplicity, trading session market hours of the New York Stock Exchange (NYSE), London Stock Exchange (LSE) and Tokyo Stock Exchange (TSE) were used in the feature construction process, instead of indexes' returns volatility, unlike the work of [Liu and Serletis \(2019\)](#) where the authors investigated the spillover effects of equities indexes RV on a basket of cryptocurrencies using GARCH-M. Trading sessions are defined as market hours during which a given exchange is open. Note that we considered extended versions of trading sessions, *i.e.* trading sessions start 30 minutes before the opening hour, and end 30 minutes after the closing hour for each market in this paper. Moreover, daylight saving (DST) was accounted for in our experiments. These three equity exchanges were chosen as their combination covers as many equity trading hours as possible. Given their distinct location and little overlap in trading hours, we shall refer to these three trading venues subsequently as follows: US for NYSE, EU for LSE and ASIA for TSE. HAR_{eq} dynamics are captured by the following model

$$\begin{aligned} RV_{i,t+u}^{(u)} = & \alpha + \beta^{(u)} RV_{i,t}^{(u)} + \beta^{(1H)} RV_{i,t}^{(1H)} + \beta^{(6H)} RV_{i,t}^{(6H)} \\ & + \beta^{(12H)} RV_{i,t}^{(12H)} + \beta^{(1d)} RV_{i,t}^{(1d)} \\ & + \beta^{(1w)} \overline{RV_{i,t}^{(1w)}} + \beta^{(1m)} \overline{RV_{i,t}^{(1m)}} + \beta^{(6m)} \overline{RV_{i,t}^{(6m)}} \\ & + \beta^{(US)} US_{t+u} + \beta^{(EU)} EU_{t+u} + \beta^{(ASIA)} ASIA_{t+u} + \epsilon_{t+u} \end{aligned} \quad (7)$$

Eqn (7) includes three new features in comparison to the HAR base model in Eqn (6). In addition to the aforementioned exchange trading sessions, we used VIX intraday RV dynamics to forecast 1D-ahead crypto intraday RV. ProShares VIX Mid-Term Futures ETF (VIXM hereafter) data from LOBSTER database was chosen as a proxy to this end. Since VIXM is not traded 24/7 in contrast to crypto pairs, we scaled VIXM RV to accommodate for overnight effects as described by [Hansen](#)

(2005). We leveraged the full orderbook (10 levels) with ML techniques allowing for dimensionality reduction while we restricted VIXM RV to the top of the book for simpler methods, *i.e.* Linear Regression (see section 4.4.1).

$$\begin{aligned}
RV_{i,t+u}^{(u)} = & \alpha + \beta^{(u)} RV_{i,t}^{(u)} + \beta^{(1H)} RV_{i,t}^{(1H)} + \beta^{(6H)} RV_{i,t}^{(6H)} \\
& + \beta^{(12H)} RV_{i,t}^{(12H)} + \beta^{(d)} RV_{i,t}^{(1d)} \\
& + \beta^{(1w)} \overline{RV_{i,t}^{(1w)}} + \beta^{(1m)} \overline{RV_{i,t}^{(1m)}} + \beta^{(6m)} \overline{RV_{i,t}^{(6m)}} \\
& + \sum_{l \in L^D} \sum_{k=1}^K RV_{VIXM,k,t}^{(l)} + \sum_{l \in L^S} \sum_{k=1}^K \overline{RV_{VIXM,k,t}^{(l)}} + \epsilon_{t+u}
\end{aligned} \tag{8}$$

where K represented the number of VIXM orderbook levels and is either equal to 1 or 10 according to the ML technique implemented. We shall refer to Eqns (7) and (8) as $\mathbf{HAR}_{eq}$ and $\mathbf{HAR}_{eq}^{VIXM}$ respectively in the remaining parts of this work.

4.2.5 Vector autoregressive

Given an asset i in our crypto universe, one can accommodate AR model to incorporate other crypto pairs' RV to model dynamics and form a system of equations to represent crypto RVs simultaneously. Resorting to vector autoregressive (VAR) model is one of the numerous ways to achieve this. VAR is described by Greene (2008) as follows

$$\begin{aligned}
\mathbf{RV}_{t+u}^{(u)} &= \mathbf{A} + \mathbf{B}^{(u)} \mathbf{RV}_t^{(u)} + \mathbf{E}_{t+u}^{(u)}, \\
\mathbf{RV}_{t+u}^{(u)} &= \left[RV_{1,t+u}^{(u)}, RV_{2,t+u}^{(u)}, \dots, RV_{N,t+u}^{(u)} \right]^\top, \\
\mathbf{A} &= [\alpha_1, \alpha_2, \dots, \alpha_N]^\top, \\
\mathbf{B}^{(u)} &= \begin{bmatrix} \beta_{(1,1)}^{(u)} & \beta_{(1,2)}^{(u)} & \dots & \beta_{(1,N)}^{(u)} \\ \beta_{(2,1)}^{(u)} & \beta_{(2,2)}^{(u)} & \dots & \beta_{(2,N)}^{(u)} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{(N,1)}^{(u)} & \beta_{(N,2)}^{(u)} & \dots & \beta_{(N,N)}^{(u)} \end{bmatrix}, \\
\mathbf{RV}_t^{(u)} &= \left[RV_{1,t}^{(u)}, RV_{2,t}^{(u)}, \dots, RV_{N,t}^{(u)} \right]^\top, \\
\mathbf{E}_{t+u}^{(u)} &= \left[\epsilon_{1,t+u}^{(u)}, \epsilon_{2,t+u}^{(u)}, \dots, \epsilon_{N,t+u}^{(u)} \right]^\top.
\end{aligned} \tag{9}$$

4.3 Training schemes

Motivated by the presence of commonality shown in Section 3.2, we trained the models in Section 4.2 under different schemes. We define training schemes as different ways to incorporate commonality in our models. The following schemes are considered

- **Single Asset Model (SAM)** refers to the routine in which each RV forecast was computed using a pair's own features for a given model defined in Section 4.2.

- **Clustered Asset Model (ClustAM)** describes the scheme where features of all cluster members, *i.e.* neighbours, are incorporated in the models. This requires clustering techniques to be applied on the data beforehand to allocate each pair to a cluster. This scheme allows one to augment models trained under **SAM** and include commonality in the models with neighbours' features. We describe the selection procedure of the number of clusters and the clustering methodology itself in Section 4.3.1.
- **Cross-Asset Model (CAM)** characterizes the setting in which a pair's RV forecast was computed using features from the entire universe of token pairs. Given the large amount of individual features, we rely on various regularization methods mentioned in Section 4.4.2 to train the models.
- The last layer of commonality is introduced in the **Universal Asset Model (UAM)** training scheme. Inspired by universal models in the literature (see Engle and Sokalska (2012) and Bollerslev et al. (2018)), we pooled data after computing features for individual pairs. We make use of one hot encoding to handle panel data. Given the large amount of observations resulting from pooling, DL based techniques offer strong alternatives and we shall utilize one of them: Long Short-Term Memory (LSTM). Additionally to LSTM, we fitted a vector autoregressive (VAR). These regression methods are briefly described in Sections 4.5 and 4.2.5 respectively.

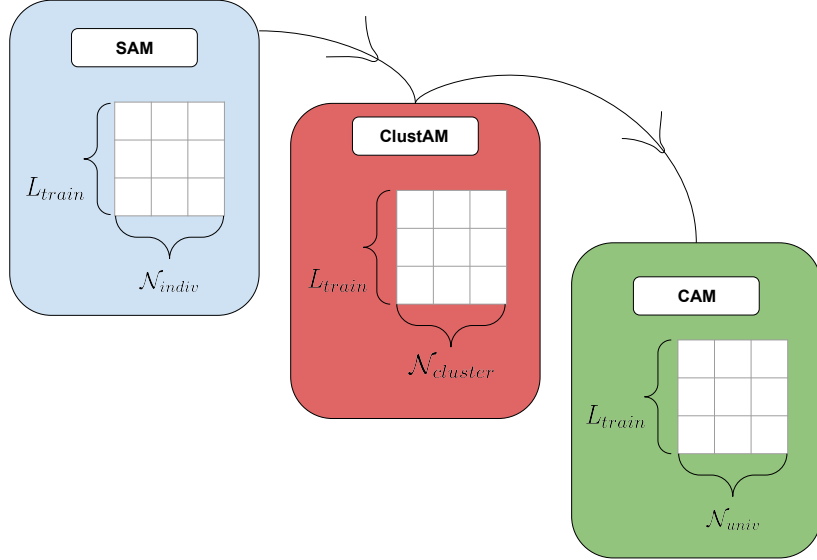


Figure 6: Dimensions of training data under three training schemes.

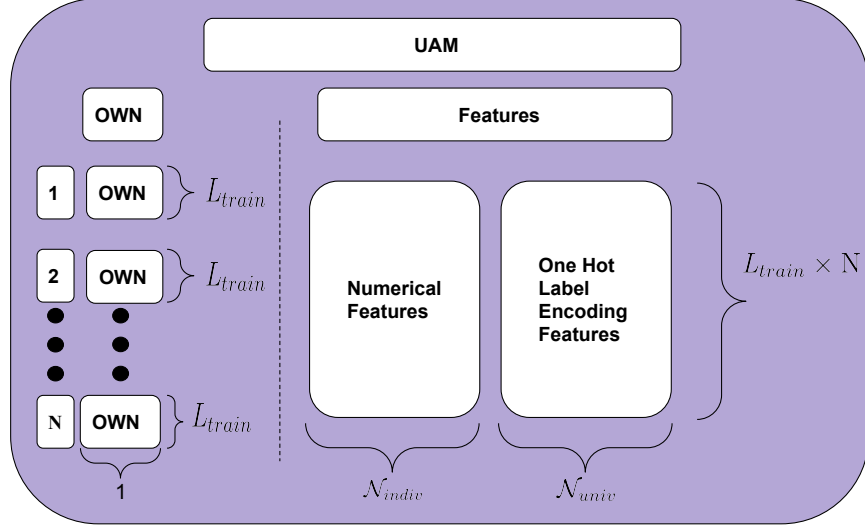


Figure 7: Dimensions of training data under **UAM** training scheme.

Figure 6 illustrates how the feature dimensions change according to the training scheme considered. The amount of observations is not impacted by the training scheme. Nevertheless, the number of features increases as more commonality is allowed within the models (indicated by the directed curved arrow in Figure 6). The different feature sizes are:

- $\mathcal{N}_{indiv}$: corresponds to the number of features of each model introduced in Section 4.2; this number is model specific.
- $\mathcal{N}_{cluster}$: let C_i refer to the i -th cluster and $\mathcal{N}_{C_i}$ to the size of that cluster. The set of features is composed of individual features per neighbor. In other words, $\mathcal{N}_{cluster} = \mathcal{N}_{indiv} \times \mathcal{N}_{C_i}$. The first layer of commonality is introduced in the model through the information contained in the cluster.
- $\mathcal{N}_{univ}$: let N denotes the size of our cryptocurrency universe. The number of features in **CAM** is $\mathcal{N}_{univ} = N \times \mathcal{N}_{indiv}$. This can be considered as a special case of **ClustAM** where there is only one cluster, the whole universe. The second commonality is captured through incorporating the individual features of each cryptocurrency pair.

4.3.1 Clustering

In preparation for model training under the **ClustAM** setting, clustering methods were applied to the data. This set of techniques allow one to partition data into distinct groups such that the intra-group difference is minimized, while the inter-group difference is maximized. In this study, we made use of the classical K-Means algorithm to solve this task. K-Means is a simple unsupervised learning method designed to split data into K distinct and non-overlapping clusters. The implementation of this algorithm requires one to choose the number of clusters, K , beforehand.

The selection procedure, that we followed, is similar to one of the approaches adopted by [Lúcio and Caiado \(2022\)](#). We trained the K-Means algorithm for values of K ranging from 2 to $N-1$, where N is the size of our cryptocurrency universe. Note that these two bounds were not chosen

randomly as $K = 1$ and $K = N$ are equivalent to **CAM** and **SAM** respectively. The selection criterion is based on *Silhouette*. The *Silhouette* score was developed by [Rousseeuw \(1987\)](#), and measures object similarity to their own cluster versus other clusters. It lies within $[-1, 1]$ and a value closer to 1 is desirable. Figure 8 displays *Silhouette* scores for the values that we tested to select the optimal K . Figure 8 suggests that the optimal value is $K = 2$, hence the data was partitioned into two groups: DOGE and the remaining crypto pairs in our universe. This is due to DOGE’s status as a meme coin, while the rest of our universe do not belong to that category

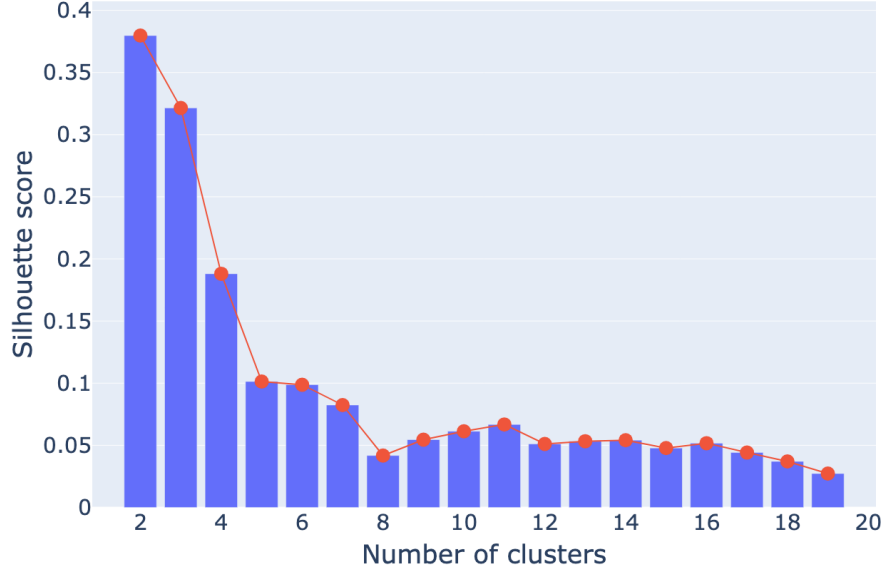


Figure 8: Silhouette score versus number of clusters.

4.4 ML techniques

We briefly summarize all ML algorithms used to train the models described in Section 4.2. Given the focus of our paper, these algorithms are not expanded on thoroughly, and the reader is referred to [James et al. \(2013\)](#) for further explanations.

4.4.1 Linear Regression (LR)

For any pair $(\mathbf{X}, \mathbf{Y})$, where $\mathbf{X}$ is a $T \times p$ feature matrix⁵ and $\mathbf{Y}$ is a $T \times 1$ target vector, LR minimizes the squared L_2 -norm of the difference of $\mathbf{Y}$ and the fitted values, $\hat{\mathbf{Y}} = \mathbf{X}\beta$.

Since **SAM** restricts commonality from RV dynamics, ergo limiting the number of features, this regression technique is only applied under this training scheme. For the remaining training schemes, we applied regularization methods.

4.4.2 Regularization

When the number of features increases, as it usually the case in big data, one is exposed to the curse of dimensionality. Plenty of workarounds exist to tackle this problem and regularization, or

⁵ T and p are the number of observations and features respectively.

shrinkage, is one of them. LASSO and Ridge belong to this class of regression techniques. The estimated coefficients in Ridge are shrunk towards zero relative to their LR counterparts during the optimization process. Note that while they are shrunk towards zero, they never attained zero. This contributes to variance reduction. LASSO can eliminate coefficients with zero-estimates. These techniques we used in our experiments supply the LR objective function with an additional shrinkage penalty term.

4.4.3 Principal Component Regression (PCR)

Another approach to perform dimensionality reduction relies on Principal Component Analysis (PCA). PCA consists of projecting the input data onto a lower-dimensional space of dimension $\mathcal{N}_{pc}$, where $\mathcal{N}_{pc}$ is the number of principal components that explain most of the data variation. In this work, we selected $\mathcal{N}_{pc}$ such that 90% of the data variance is explained. Once the $\mathcal{N}_{pc}$ first components have been computed, they are used as features in LR. Under commonality schemes, namely **ClustAM**, and **CAM**, we extended the design matrix formed by the selected principal components by adding model features specific to a given pair, as Figure 9 suggests. PCR refers to the process of using principal components instead of features in LR, and we refer to this regression as such for both non-commonality (**SAM**) and commonality schemes hereafter.

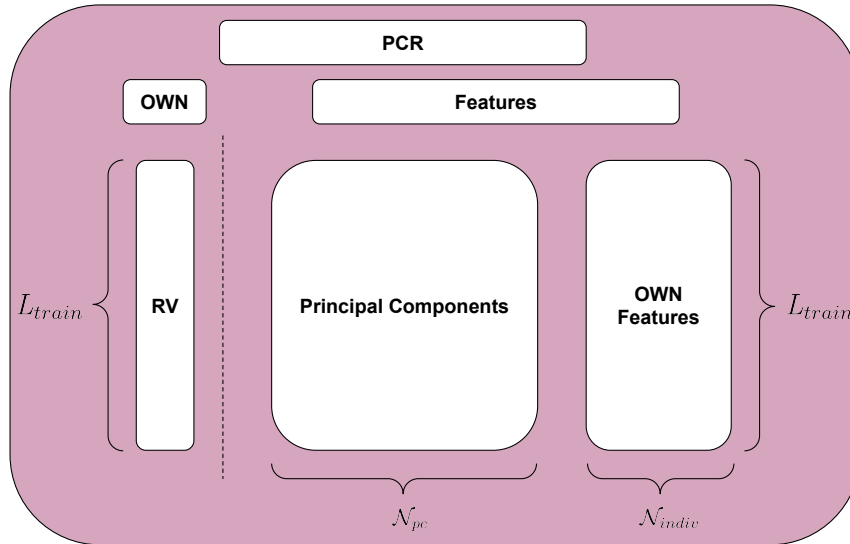


Figure 9: Target and PCR features.

4.4.4 Boosted trees

The aforementioned ML techniques perform well in scenarios where the assumption of linearity between the target and the features holds. However, this is violated more often than not. Decision trees offer capabilities to capture nonlinear relationships. While they are easily interpretable, out-of sample performance of decision trees is usually weak, hence the weak learner terminology used to name them. One way to enhance their performance is through boosting. In boosting, weak learners are trained sequentially, *i.e.* each tree uses information from previously trained trees and is fed with a modified version of the input features (see [James et al. \(2013\)](#)). In our experiments,

we used the LightGBM algorithm to train our tree-based models given its fast and efficient training and we made use of Optuna ([Akiba et al. \(2019\)](#)) for hyperparameter optimization.

4.5 DL technique

We briefly present the DL technique we employ in our study and provide a comprehensive overview of the DL model architecture used in our experiments in this section.

4.5.1 Long Short-Term Memory (LSTM)

LSTM was introduced by [Hochreiter and Schmidhuber \(1997\)](#) and have been applied to numerous RV forecasting problems, *e.g.* [Lin et al. \(2022\)](#). Its architecture has been favored as it allows one to retain past information of time series to forecast future values similarly to recurrent neural networks (RNN) thank to the presence of feedback connections in this family of DL methodologies. However, LSTM allows one to circumvent the exploding/vanishing gradient shortfall intrinsically linked to RNN architecture as the number of layers increases in RNN. [Chen and Ge \(2019\)](#) offer an elaborated comparison between RNN and LSTM for any avid readers keen on improving their comprehension of both DL techniques.

In this work, the hyperparameter space permits to train LSTM-based models as simple as single-LSTM block models with a linear final layer to multi-LSTM block models with a multi-layer perceptron (MLP). A detailed overview of the hyperparameter tuning of this model is provided in [Appendix A](#).

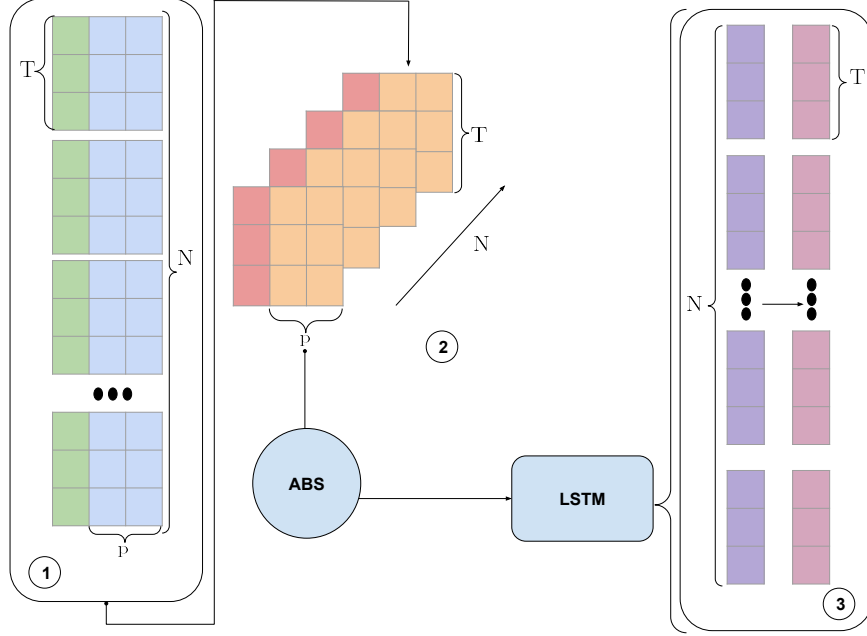


Figure 10: LSTM flow (T: number of observations per pair; N: number of pairs; p: amount of features). Individual target and features are pooled in step 1. Data is rearranged into tensors where both target (green) and features (blue) have been transformed (red and orange respectively). Given the static nature of L^S , we opted for standardizing the data by diving each feature by its maximum value in absolute value term. Note that this operation is performed for each pair separately. This altered dataset is fed to our LSTM model. This is whole sequence of operations constitute the second step of the LSTM flow. Once raw output (purple) has been generated, it is unfolded and rescaled back to the original scale (lilac). Step 3 captures the last part of the procedure.

5 Performance evaluation

5.1 Model performance

This section covers our evaluation procedure which was performed on rolling out-of-sample tests and is threefold:

- We report Quasi-likelihood (QLIKE) scores in line with the volatility forecast literature (see [Patton \(2011\)](#); [Zhang et al. \(2023\)](#))

$$QLIKE = \frac{1}{N} \sum_{i=1}^N \frac{1}{\#Test} \sum_{t=1}^{\#Test} \frac{RV_{i,t}^{(h)}}{\widehat{RV_{i,t}^{(h)}}} - \log \left(\frac{RV_{i,t}^{(h)}}{\widehat{RV_{i,t}^{(h)}}} \right) - 1. \quad (10)$$

The QLIKE superiority in volatility forecasting problems has been demonstrated by [Patton and Sheppard \(2009\)](#), compared to Mean Squared Error (MSE). For this reason, we shall emphasize QLIKE over MSE in our assessment of the model forecasts. The best performing models per L_{train} are shown in Table 2.

• In addition, we computed the pairwise Diebold-Mariano (DM) tests from our experiments. This pairwise test was introduced by [Diebold and Mariano \(1995\)](#) and allows for relative comparison of cross-sectional mean of forecast errors of two models. For any pair of models, denoted as 1 and 2, the DM test statistics are given by

$$DM_{12} = \frac{\bar{d}_{12}}{\hat{\sigma}_{12}} \quad (11)$$

$$d_{12,t} = \frac{1}{N} \sum_{t=1}^N \left(\left(\hat{e}_t^{(1)} \right)^2 - \left(\hat{e}_t^{(2)} \right)^2 \right), \quad (12)$$

$\hat{e}_t^{(i)}$ corresponds to the forecast error for the i -th model at time t . The numerator $\bar{d}_{12}$ and the denominator $\hat{\sigma}_{12}$ denote consistent estimators of the mean and the standard deviation of $d_{12,t}$ respectively. In this test, $DM_{12} \leq 0$ indicates that model 1 outperforms model 2, where $DM_{12} \geq 0$ implies that model 2 dominates model 1. The DM test results of the best models selected versus all the models trained in **SAM** are displayed in [Appendix B](#).

• We segmented time series into three market regimes to evaluate the performance of our models. These market regimes were defined based on contemporaneous market volatility values (see [Eqn 3](#)). These values were resampled on a daily frequency and sorted within the following three bins, based on percentile: $(0, 45^{th} \text{ perc.}]$, $(45^{th} \text{ perc.}, 90^{th} \text{ perc.}]$, and $(90^{th} \text{ perc.}, 100^{th} \text{ perc.}]$. We shall refer to them as *low*, *normal*, and *high* market regimes hereafter. The best performing models per market regime are displayed in [Table 2](#).

	L_{train}			vol regime		
	$1w$	$1m$	$6m$	<i>low</i>	<i>normal</i>	<i>high</i>
Model	AR ^(CAM,1w,LightGBM)	AR ^(CAM,1w,LightGBM)	HAR ^(CAM,6m,VIXM) _{EQ}	AR ^(UAM,1m,LightGBM)	HAR ^(UAM,1m,LightGBM)	HAR ^(CAM,6m,VIXM) _{EQ}
QLIKE	1.11	0.75	0.76	0.57	0.80	0.84

Table 2: Best performing model per L_{train} and best performing model per volatility regime. This table reports each model with their respective QLIKE score.

[Table 2](#) hints that allowing intraday RV commonality in our modeling framework enhances model performance across all volatility regimes. As market volatility increases, our framework yields better forecasts. As suggested by [Table 2](#), the presence of intraday RV commonality in model dynamics contributes to higher model performance for L_{train} values greater than $1w$. Nevertheless, modeling individual RV dynamics leads to a lower QLIKE for $L_{train} = 1w$. These models will be referred to $M^{L_{train}}$ for a given L_{train} hereafter.

5.2 Connectivity in the crypto volatility network

In addition to the model performance evaluation, we conducted a spillover effect analysis of the best performing model for each L_{train} window. Similarly to [Iyer and Popescu \(2023\)](#), we made use of [Diebold and Yilmaz \(2012\)](#) framework to analyze RV spillover by employing forecasts produced by M^{1w} , M^{1m} , and M^{6m} . We report the results of our spillover analysis only for the L_{train} values

associated with the lowest QLIKE values, *i.e.* $1m$, for sake of brevity. Nevertheless, the outcome of our analysis for the remaining L_{train} is shown in the appendix section C. Figures 11 and 13 display RV spillover within our crypto networks and the total spillover index (TSI).

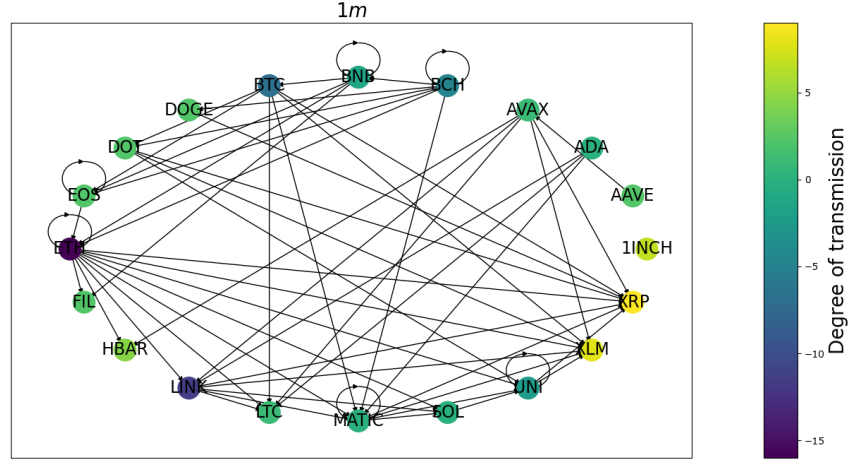


Figure 11: Connectivity network. Each crypto pair is represented as a node within our network, while the edges indicate a directed relationship between two pairs among the top 25th spillover percentile. The color of each node indicates whether a pair is a net transmitter or receiver within our network. This is dictated by the node balance, which is defined as the difference between the out-degree and the in-degree of node i . If the balance of a given node is positive (resp. negative), that node is labeled as a transmitter (resp. receiver).

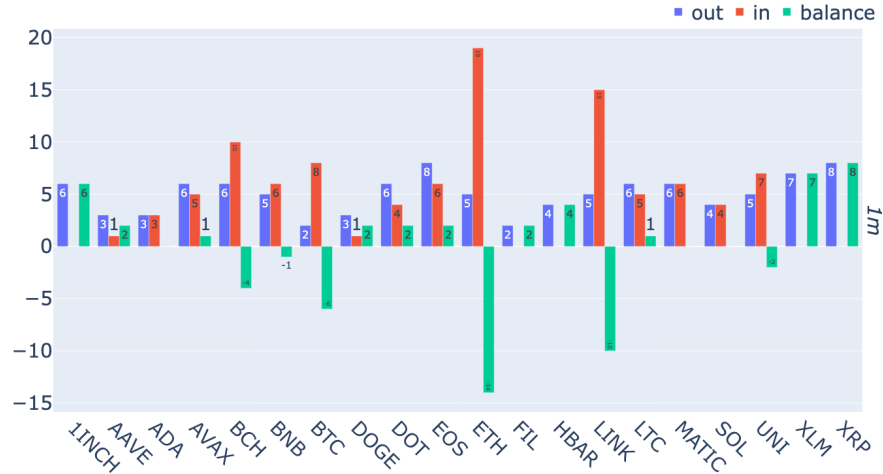


Figure 12: Nodes: degrees and balance - $1m$. To complement Figure 11, the degrees and the balance of each node in the network are reported in this figure.

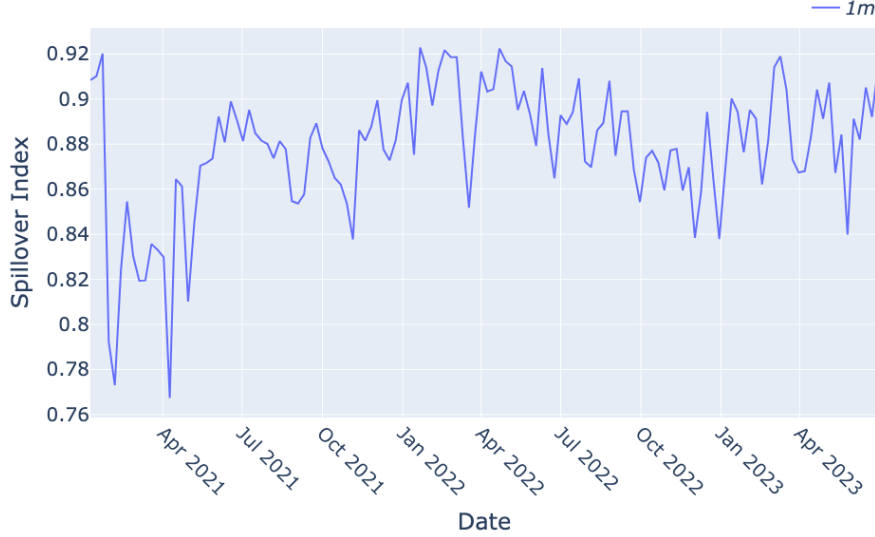


Figure 13: TSI - 1m. The amount of variation in RV explained within our network oscillates between 0.77 and 0.92 as suggested by this figure.

Figures 11 and 12 provide additional insights into the structure of our network. Firstly, one can clearly identify ETH as the pair with the lowest balance in the former figure. In other words, ETH is the pair whose RV is influenced the most by other neighbors within the top 25th percentile edges. Secondly, the benefits of commonality in intraday crypto RV is further showed by Figure 13. Indeed, 7-day TSI average is above 0.77 throughout the time period considered. As result of this, one may claim that a significant part of RV variation is captured within our network.

6 Economic benefits

In this section, we present the study of a simple trading strategy based on RV forecasts derived with the model yielding the lowest QLIKE score. Furthermore, we compare the performance of that trading strategy to one benchmark.

6.1 Trading strategy

Encouraged by the benefits of commonality empirically shown in Section 5, we investigated the economic benefits of using commonality in forecasting 1D-ahead RV through the backtesting of a trading strategy. Guo (2000) tested two trading strategies on dollar/mark options leveraging volatility forecasts: a Delta-Neutral Trading strategy, where one is hedged against directional risk by holding the underlying in addition to a position in an option, and a Straddle Trading strategy. Inspired by the latter, we shall make use of our RV forecasts to buy straddles when RV increases or sell straddles otherwise. Indeed, Natenberg (1994) describes straddles as appropriate portfolios to trade volatility variations. They are composed of a call and a put options with the same strike and maturity. Similarly to price data mentioned in Section 3, the option chain data used in this paper was fetched from Tardis and all contracts in this data set were traded on Deribit⁶. We opted for

⁶<https://www.deribit/>

ETH contracts⁷ to implement that trading strategy on since ETH is a consistent net transmitter in our RV crypto network as shown in Figure 17. The implementation of our straddle-based trading strategy is summarized as follows:

- All straddles available are bought or sold at time t according to signals generated at time $t-I$.
- *Deribit* offers a large variety of ETH options. For any day d , one can trade the following expiration: $d+1$, $d+2$, $d+3$, next week, in two weeks, in three weeks, end of next month, in two months, in five months, in six months, and in twelve months. Note that the last Fridays of the aforementioned period are the default expiry days (08:00AM UTC) unless the period was listed in days. Furthermore, the range of strikes on which the contingents are written on covers the range from $.8S_0$ to $1.5S_0$ approximately. This implies that our whole option universe consists of thousands of contingents. However, only a subset of these contracts are quoted on day d and the composition of straddles, *i.e.* both call and put legs should be available on the same strike and the same maturity at the same time, restricts us considerably in terms of options one can trade at time t . Readers are referred to *Deribit* for more insights on *Deribit*'s pricing methodology.

6.2 Backtesting results

Our trading strategies are compared against one benchmark:

- Trading strategies that rely on signals generated by three GARCH(1,1) models that were fitted for each L_{train} . GARCH(1,1) was selected given its status in volatility forecasting literature. These strategies will be identified as GARCH when we refer to these strategies as a group and $GARCH^{L_{train}}$ for a specific L_{train} value henceforward.

Transaction fees (TC) advised on *Deribit*, *i.e.* 3 bps of ETH, are taken into consideration in our backtesting as displayed in Eqns (15) and (16). We report Sharpe ratio (SR) and cumulative P&L

⁷Note that ETH contracts are settled in ETH on *Deribit*. In order to facilitate reporting and computations, we used ETH spot level to convert ETH option prices in USDT.

curves as part of the backtesting. The detail of our backtesting methodology is given here below:

$$signals_t = (\widehat{RV}_t - RV_{t-h}) \cdot \mathbb{1}_{\{|filter_t| \leq \delta\}}, \quad (13)$$

$$filter_t = \frac{\widehat{RV}_t - RV_{t-h}}{\widehat{RV}_t^{L_{train}}}, \quad (14)$$

$$ret_t^{straddle,buy} = \frac{1}{N} \sum_{\mathcal{N}^{straddles}} \frac{straddle_{i,t} - TC_t - straddle_{i,t-h} - TC_{t-h}}{straddle_{i,t}} \cdot \mathbb{1}_{\{signals_t > 0\}} \cdot b_{i,t}, \quad (15)$$

$$ret_t^{straddle,short} = \frac{1}{N} \sum_{\mathcal{N}^{straddles}} \frac{straddle_{i,t-h} - TC_{t-h} - straddle_{i,t} - TC_t}{straddle_{i,t}} \cdot \mathbb{1}_{\{signals_t < 0\}} \cdot b_{i,t}, \quad (16)$$

$$ret_t^{straddle} = (ret_t^{straddle,buy} + ret_t^{straddle,short}) / 2, \quad (17)$$

$$P\&L_t = \sum_{t=1}^T ret_t^{straddle}, \quad (18)$$

$$(19)$$

where Eqns (13) to (16) were computed by [Andrada-Félix et al. \(2016\)](#). Eqn (14) is used to discard the noisiest trading signals triggered by Eqn (13). One trades on buy signals if the excess of RV forecast at time $t+h$ and RV at time t over the mean of RV computed over L_{train} exceeds a predefined threshold, δ (see [Andrada-Félix et al. \(2016\)](#)). Conversely, short signals are traded on if the excess of RV forecast at time $t+h$ and RV at time t over the mean of RV computed over L_{train} is less than $-\delta$. In this work, δ is set to 5. The size of each straddle is fixed to one in our trading strategies as mentioned in Section 6.1. As a result, we either trade one straddle per transaction or do not trade at all, i.e $b_{i,t} \in \{0, 1\}$. Figures 14, and 15 display the SR, and P&L respectively.

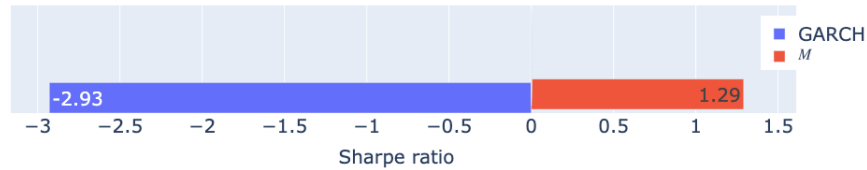


Figure 14: Annualized SR - Im . As suggested by this figure, M leads to positive risk-adjusted score while GARCH based signals yield a negative SR.

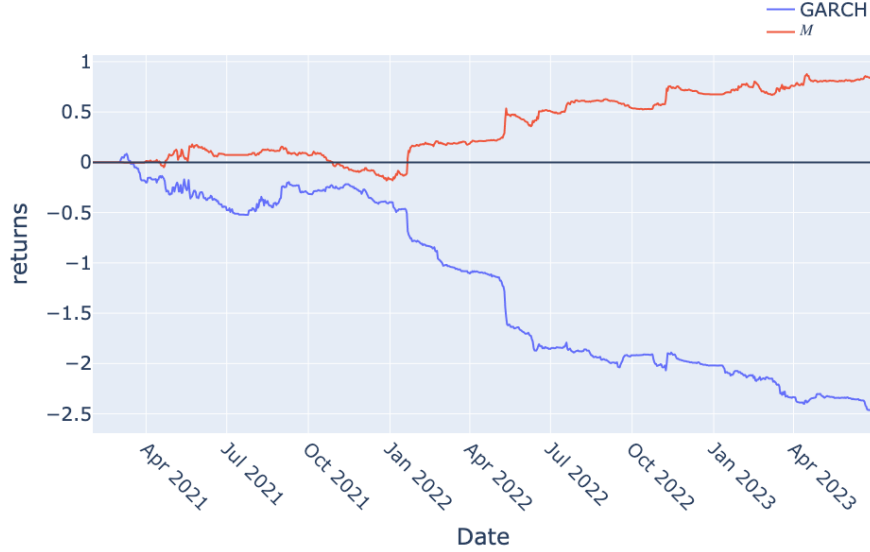


Figure 15: Cumulative P&L curves - $1m$. M^{1m} P&L curve dominates $GARCH^{1m}$ curve throughout the entire backtesting period. The first strategy reaches 82% returns and the second strategy performed significantly worse (-250%).

Figures 14 and 15 confirm that M based signals strategy outperforms both GARCH strategy in risk-adjusted and monetary terms. Moreover, our strategy tends to identify short trades more often than its GARCH counterpart as the following numbers suggest: 11369 short and 57 long signals for M^{1m} versus 10046 short and 6163 long signals for GARCH. These findings are further confirmed by the backtesting of our trading strategy for the other L_{train} values as shown in Appendix C.

7 Conclusion

This study demonstrated the benefits of intraday RV commonality in forecasting 1D-ahead intraday RV in cryptocurrency markets. This phenomenon was investigated over three lookback windows and four training schemes were defined, namely Single Asset Model, Clustered Asset Model, Cross-Asset Model, and Universal Asset Model to allow commonality to be gradually incorporated into our models. We trained a series of models using various machine learning, including deep learning, techniques and performed testing following the walk-forward method. Performance metrics were reported to evaluate model forecasting abilities. Moreover, we performed a spillover analysis within our crypto RV universe and established the presence of persistent net transmitters. Our findings were utilized in the development of a trading strategy in the nascent crypto options market in order to trade volatility. The results of our backtesting demonstrated that our models yield positive P&L over the best performing model rolling windows and outperformed the benchmark model. Supported by these results, we conclude that intraday RV commonality in cryptocurrency markets improves 1D-ahead RV forecasts and contributes to enhancing trading performance.

Several research directions can be explored following our results. Our framework can be compared with High frEQUENCY bAsed VolatilitY (HEAVY) models and regression can be performed

using other DL architecture, *i.e.* WaveNet. Examining the source of commonality in cryptocurrencies can be investigated with high-order objects like tensors, as commonality can originate from different aspects of the data, e.g., commonality across time-of-day buckets or day-of-the-month. Limit order book features, as well as on-chain based features can help increase model forecasting performance. Further analysis on different time horizons can also be conducted as well as on the benefits of commonality in covariance matrix forecasting in crypto markets. Lastly, exploring the risk premium and the term structure in cryptocurrency markets using our framework can help increase our understanding of the nascent crypto option market dynamics, and improve pricing models for crypto options.

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Appendix

A Hyperparameter Tuning

We list the parameter space that we used in the training procedure in Table 3

Training scheme	Regression	Hyperparameter space
SAM ClustAM CAM	LASSO	$\{\alpha \in [.0001, .1]\}$
SAM ClustAM CAM	Elastic	$\{\alpha \in [.0001, .1], \lambda_1 \in [.01, .99]\}$
SAM ClustAM CAM	LightGBM	$\{D \in (1, 3), \lambda \in [.01, .05, .1], f \in [.1, .2, .3, .4, .5, .6, .7, .8]\}$
UAM	LightGBM	$\{D \in (1, 3), \lambda \in [.01, .05, .1]\}$
UAM	LSTM	$\{H^S \in [2^3, 2^4, 2^5],$ $lr \in [.0001, .001, .005],$ $\#H^S \in [1, 2, 3, 4, 5],$ $\Lambda \in [0, 1],$ $\Omega_{\Lambda=1} \in [2^3, 2^4, 2^5],$ $\#\Omega_{\Lambda=1} \in [1, 2, 3],$ $d \in [.1, .11, .12, .13, .14, .15, .16, .17, .18, .19, .2]\}$

Table 3: Parameter space that we used in the training procedure presented in Section 4.

- α : factor multiplying penalty terms in LASSO and Elastic objective functions,
- λ_1 : weight allocated to L_1 penalty term in Elastic objective function. Note that Elastic objective function allows for an L_2 penalty term as well with a weight equal to $1 - \lambda_1$,
- D : maximum depth of each decision tree trained using LightGBM,
- λ : learning rate used in LightGBM training process,
- H^S : hyperparameter controlling the dimensions of the hidden state used in LSTM architecture,

- lr : learning rate used in LSTM training process,
- $\#H^S$: number of LSMT block used in the LSTM architecture,
- Λ : binary hyperparameter that allows the final layer to be either linear (0) or an MLP (1),
- $\Omega_{\Lambda=1}$: size of each layer in MLP should LSTM model permits for a non-linear final layer,
- $\#\Omega_{\Lambda=1}$: number of layers in MLP,
- d : dropout rate in LSTM model.

B Diebold-Mariano Test

			M^{1w}	M^{1m}	M^{6m}
SAM	AR	LR	-1.585774	-2.273148	-1.579194
		HAR	-1.585849	-2.273178	-1.579289
		Elastic	-1.585801	-2.273211	-1.579189
		LASSO	-1.585801	-2.273211	-1.579189
		LR	-1.585714	-2.213363	-1.579191
		LightGBM	-1.585762	-2.273002	-1.579221
		Ridge	-1.585801	-2.283612	-1.579227
	HAR _{eq}	Elastic	-1.585915	-2.273151	-1.579207
		LASSO	-1.585865	-2.273134	-1.579226
		LR	-1.585747	-2.2732	-1.579141
		LightGBM	-1.585807	-2.273077	-1.579239
		Ridge	-1.585838	-2.273222	-1.579284
	HAR _{eq} ^{VIXM}	Elastic	-1.585831	-2.288651	-1.579235
		LASSO	-1.585834	-2.273231	-1.582569
		LR	0.030343	-2.27263	-1.5792
		LightGBM	-1.585796	-2.27303	-1.579211
		Ridge	-1.585837	-2.273133	-1.579235
	RiskMetrics	EWMA	-1.595708	-2.473708	-1.603834

Table 4: DM test. As hinted by this table, we report the DM test values for best performing model for each L_{train} . These test statistics are built by comparing their error forecasts with models' error forecasts trained in **SAM**.

SAM is used as training scheme benchmark as it does not allow for any degree of commonality in model dynamics. In order to guarantee a fair assessment, we compare **SAM** models with the

same L_{train} as $M^{L_{train}}$. As suggested in Table 4, our framework yields superior results regardless of L_{train} values in general. However, M^{1w} is outperformed by $HAR_{eq}^{VIXM(SAM, 1w, LR)}$.

C Performance evaluation and backtesting

We report additional material to assess the performance of our framework across the remaining L_{train} values, *i.e.* $1w$, and $6m$. Furthermore, we display the results of our backtesting for the aforementioned L_{train} values.



Figure 16: Node degrees and balance. In complement to Figure 12, the degree of each node in our network is reported in this figure.

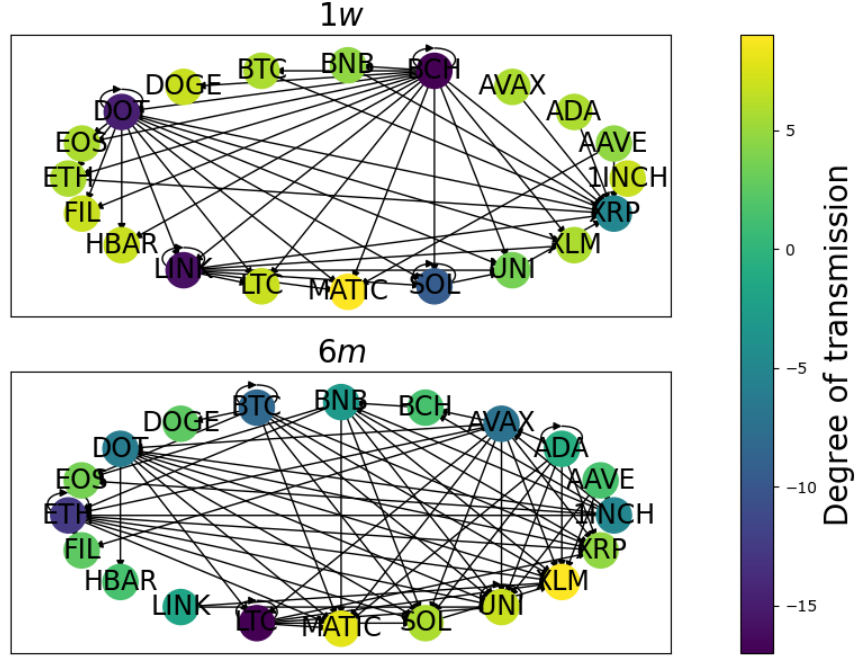


Figure 17: Connectedness network. We display the status of each pair within our network for L_{train} values not reported in Figure 11.

The receiver status of ETH for all L_{train} values strictly higher than $1w$ among the top 25th spillover percentile is corroborated in this figure.

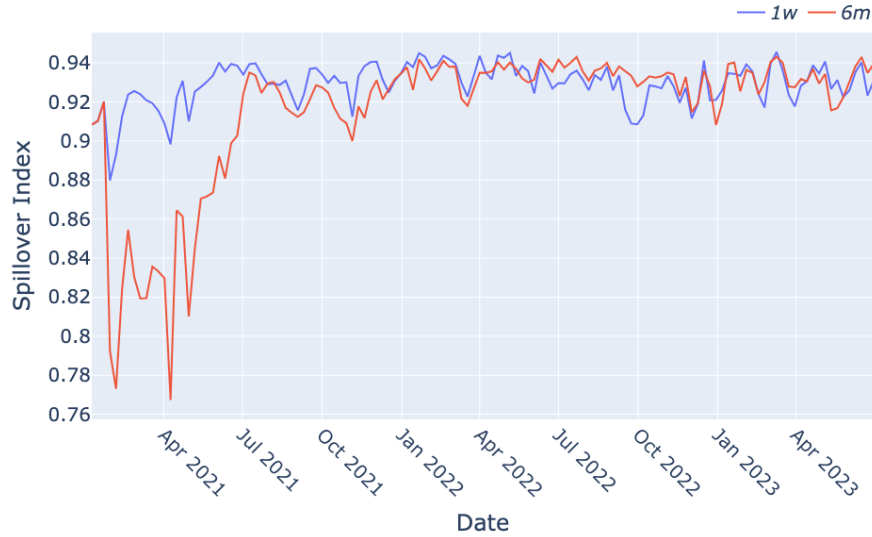


Figure 18: TSI. This figure shows the evolution of TSI for L_{train} values not reported in Figure 13.

Similarly to Figure 13, our network captures a high degree of RV variation indicated by the values reached by TSI. Indeed, TSI fluctuates approximately between .8 and .9 for both $1w$ and $6m$.

Figures 16, 17, and 18 present additional information on our crypto RV network. One can clearly identify ETH as the pair with the largest in-degree for $L_{train} \in \{1m, 6m\}$ in Figures 16 and 17. In other words, ETH is the pair for which RV is influenced the most by other neighbors within the top 25th percentile edges. Furthermore, the 7-day TSI average is consistently higher than .88 and .77 for M^{1w} , and M^{6m} respectively. This indicates that a great portion of RV variation is explained within our network.

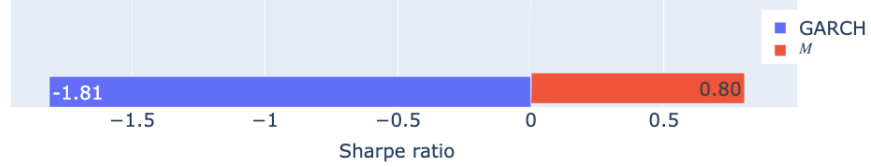


Figure 19: Annualized SR - $1w$. Both models produced a negative SR. Nevertheless, M^{1w} outperforms GARCH based trading strategy. M^{1w} generates a positive .8 SR while $GARCH^{1w}$ produced a negative SR, -1.81.

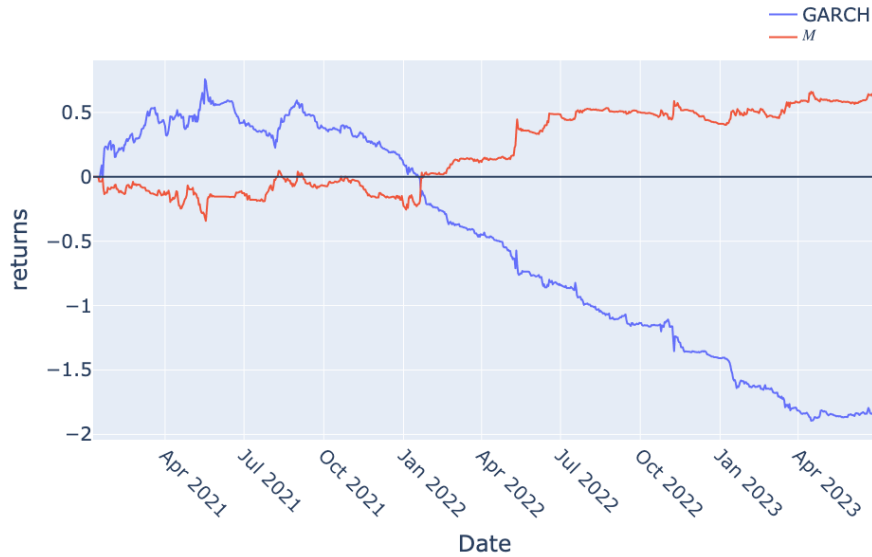


Figure 20: Cumulative P&L curves - $1w$. $GARCH^{1w}$ based signals generated positive returns for roughly one year before sinking and reaching -185% return. M^{1w} return profile grew at a steady rate after the first year of trading and it achieves 61% returns.

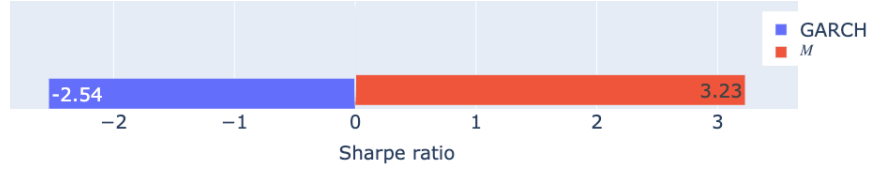


Figure 21: Annualized SR - 6m. As displayed in this figure, M^{6m} SR is positive while $GARCH^{6m}$ is negative.

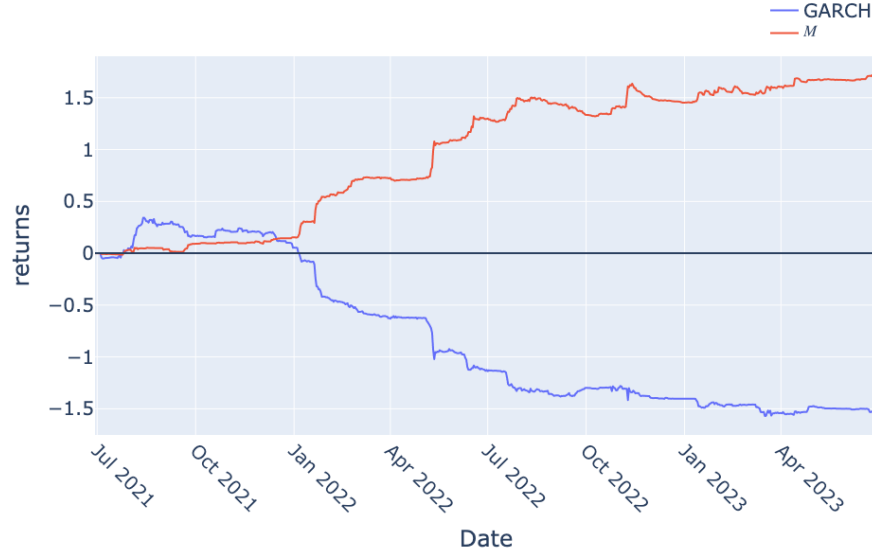


Figure 22: Cumulative P&L curves - 6m. M^{6m} P&L curve is dominated by $GARCH^{6m}$ until early 2022 and trends upwards until reaching 171% returns. $GARCH^{6m}$ P&L curve plunges and yields -155% returns.

Similarly to the backtesting results presented in Section 6.2, we provide the number of long and short signals generated by each strategy: M^{1m} and M^{6m} produced 14452 short/188 long and 7294 short/221 long signals respectively, while their corresponding GARCH models identified 10525 short/7396 long and 5039 short/4337 long opportunities.